Chelsea Zebaze

Software Engineer · Full-Stack · Data
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Summary

Software engineer with a B.S. in Mathematics from Case Western Reserve University who builds full-stack applications, backend services, and data pipelines with Python, TypeScript, SQL, FastAPI, PostgreSQL, React, and Next.js. Built PrepForward, an adaptive SAT learning platform with an OCR/ETL pipeline and layered PostgreSQL warehouse, and helped deliver a geospatial analytics platform on AWS scoring ~70,000 points across Cleveland. Comfortable across the stack, from data modeling and APIs to responsive interfaces and data analysis.

Technical Skills

- **Languages:** Python, SQL, TypeScript, R
- **Frontend:** React, Next.js, Tailwind CSS, CSS, Mapbox GL JS
- **Backend:** FastAPI, REST APIs, PostgreSQL, Supabase
- **Data Engineering:** Apache Airflow, ETL, OCR (Tesseract), data modeling, pipeline monitoring, GeoPandas
- **Data Analysis & Visualization:** pandas, matplotlib, seaborn, Microsoft Excel, Tableau
- **Machine Learning:** PyTorch, scikit-learn, TensorFlow, XGBoost, clustering, feature engineering
- **Cloud & DevOps:** AWS (Lambda, S3, API Gateway), GCP, Git, Vercel

Projects

PrepForward: **Adaptive SAT Learning Platform** — Founder & Full-Stack Developer · Dec 2025 – Present

- Built an end-to-end OCR and ETL pipeline with Tesseract to extract, clean, validate, and structure 500+ SAT questions from multiple workbook datasets.
- Designed a PostgreSQL data warehouse with Bronze, Silver, and Analytics layers for scalable ingestion, quality validation, and reporting.
- Built automated pipeline monitoring and auditing to track ETL runs, ingestion failures, OCR validation, and data quality metrics.
- Engineered analytics pipelines calculating user accuracy, time-on-task, and topic mastery to drive adaptive content sequencing.
- Orchestrated scheduled ETL and machine learning workflows with Apache Airflow to automate data processing and model retraining.

MapsModes: City-Scale Mobility Analytics Platform — Bike Accessibility Lead, 5-person capstone team · Aug 2025 – Dec 2025

- Designed a 7-factor weighted Bike Score model (lane density, intersections, mode share, amenities, hub proximity, bike availability, slope) to rate bikeability by census tract across Cleveland.
- Built a geospatial ETL pipeline in GeoPandas joining city bike-network, OpenStreetMap, Census, GTFS, GBFS, and USGS elevation data; computed slope statistics with rasterio.
- Extended the model into a route-level scoring algorithm using inverse-distance decay to grade individual bike routes.
- Integrated the model into a cloud-deployed platform (AWS Lambda, S3, API Gateway, FastAPI backend; React + Mapbox GL JS frontend) serving interactive choropleth maps across ~70,000 scored points in a public live demo.

Personal Portfolio Website — Next.js, TypeScript, Tailwind CSS · Oct 2026

- Built a responsive portfolio from a Figma design with reusable React components and a custom Tailwind theme.
- Deployed on Vercel with automatic builds on every push to GitHub.

Experience

Outlier AI — AI Response Reviewer & Content Creator · Remote · Apr 2024 – Dec 2025

- Evaluated Python, SQL, and JavaScript solutions for correctness, efficiency, and best practices in structured QA workflows for AI model training.
- Assessed AI-generated responses for accuracy and reasoning quality; documented inconsistencies, edge cases, and failure patterns to support model improvement.
- Created and refined prompts across technical and analytical domains to test reasoning, coding, and language capabilities while meeting productivity and accuracy targets.

Varsity Tutors — Contract Math & Standardized Testing Tutor · Remote · Jun 2024 – Present

- Tutor Algebra I/II, Pre-Calculus, Calculus I/II, SAT, and ACT, breaking complex problems into structured, step-by-step solutions.
- Design individualized study plans from diagnostic performance data, working with students and parents to set goals and track progress.

Education

Case Western Reserve University — B.S. in Mathematics · Cleveland, OH · Dec 2025

- Relevant coursework: Discrete Mathematics, Linear Algebra, Numerical Analysis, Scientific Computing, Statistics.
- Senior Project (CSDS 395): co-developed MapsModes, a geospatial mobility-analytics platform (see Projects).